



2018

Preliminary
Assessment
Task 2

Preliminary Mathematics Extension

General Instructions

- Reading Time – 2 minutes
- Working Time – 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 5 – 7, show relevant mathematical reasoning and/or calculations

Total marks – 40

Section I – 4 marks (pages 2 – 3)

- Attempt Questions 1 – 4
- Allow about 6 minutes for this section

Section II – 36 marks (pages 4 – 6)

- Attempt Questions 5 – 7
- Allow about 49 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER: _____

Question	1 – 2	3 – 4	5	6	7	Total
Trigonometry	/2		/6	/13	/11	/32
Geometry		/2	/6			/8
	/2	/2	/12	/13	/11	/40

Section I

4 marks

Attempt Questions 1–4

Allow about 6 minutes for this section

Use the multiple-choice answer sheet for Questions 1–4.

1 What is the exact value of 55° in radians?

(A) $\frac{180\pi}{55}$

(B) $\frac{11\pi}{36}$

(C) $\frac{36}{11\pi}$

(D) $\frac{55}{180\pi}$

2 What is the exact value of $\sin 2x$ if $\cos x = \frac{\sqrt{2}}{5}$ and x is in the 4th quadrant?

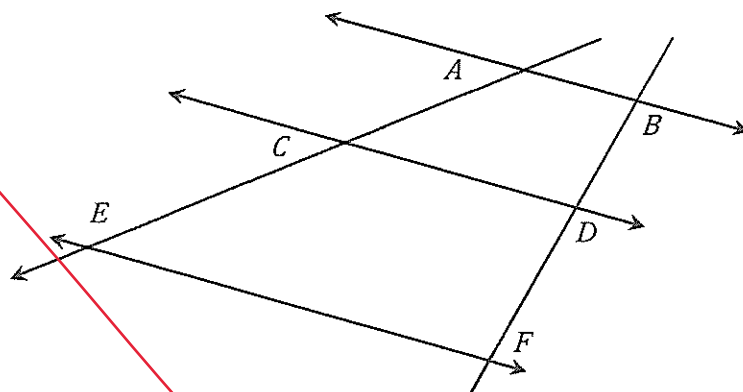
(A) $\frac{4\sqrt{23}}{25}$

(B) $\frac{2\sqrt{46}}{25}$

(C) $-\frac{2\sqrt{46}}{25}$

(D) $-\frac{4\sqrt{23}}{25}$

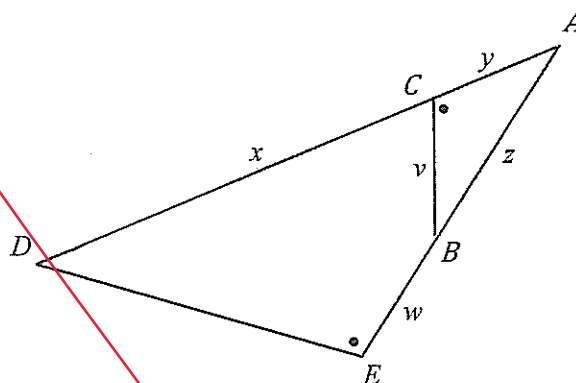
- 3 In the diagram $AB \parallel CD \parallel EF$.



Which statement is **NOT** correct?

- (A) $\frac{AC}{AE} = \frac{BD}{BF}$
 (B) $\frac{AC}{CE} = \frac{BD}{DF}$
 (C) $\frac{AC}{BD} = \frac{CE}{DF}$
 (D) $\frac{BD}{BF} = \frac{CD}{EF}$

- 4 The two triangles in the diagram below are similar.



What is the length of DE ?

- (A) $\frac{vz + vw}{z}$
 (B) $\frac{vx + vy}{z}$
 (C) $\frac{vx + vy}{y}$
 (D) $\frac{yz + yw}{v}$

Section II

36 marks

Attempt Questions 5 - 7

Allow about 49 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 5 – 7, your responses should include relevant mathematical reasoning and/or calculations.

Question 5 (12 marks) Use a SEPARATE writing booklet.

(a) Prove that $\cos(A+B) - \cos(A-B) = -2 \sin A \sin B$. 2

(b) Let $\sin x + \sqrt{3} \cos x \equiv R \sin(x + \alpha)$.

(i) By expanding the right-hand side and equating coefficients, find the values of R and α . 3

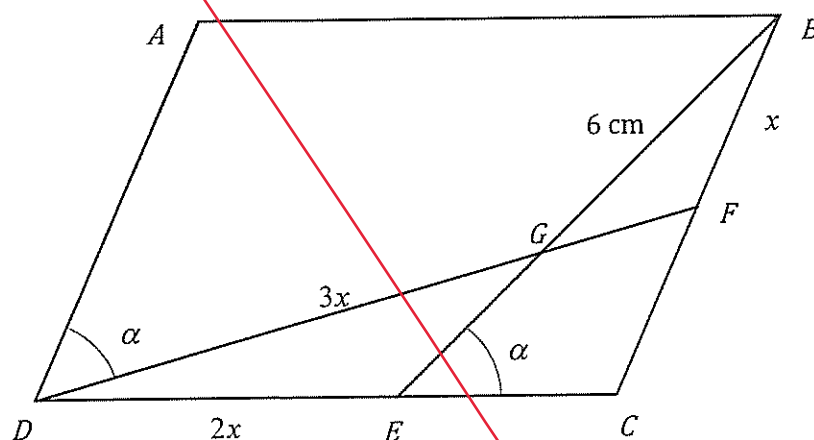
(ii) Hence, or otherwise, find the maximum value of the function 1

$$f(x) = \sin x + \sqrt{3} \cos x - 5.$$

(c) ~~$ABCD$ is a parallelogram.~~

~~F and E are two points on BC and DC respectively such that $BF = x$, $DE = 2x$ and $\angle ADF = \angle BEC = \alpha$.~~

~~FD meets BE at G such that $DG = 3x$ and $BG = 6 \text{ cm}$.~~



Copy the diagram into your writing booklet

(i) Prove that $\angle GDE = \angle GBF$. 2

(ii) Prove that $\triangle GDE$ is similar to $\triangle GBF$. 2

(iii) Hence find the value of x . 2

Question 6 (13 marks) Use a SEPARATE writing booklet.

(a) Solve the equation $\cos 2\alpha = \sin \alpha$ for $0 \leq \alpha \leq 2\pi$. 3

(b) Use the substitution $t = \tan \frac{x}{2}$ to solve 3

$$\sin x + \cos x = -1 \text{ for } 0 \leq x \leq 2\pi.$$

(c) (i) Find a general solution of $\cos 2x = \frac{1}{2}$. 2

(ii) Hence or otherwise solve $\cos 2x = \frac{1}{2}$ for $-2\pi \leq x \leq 0$. 1

(d) (i) Prove that $\cot \theta - \tan \theta = 2 \cot 2\theta$. 2

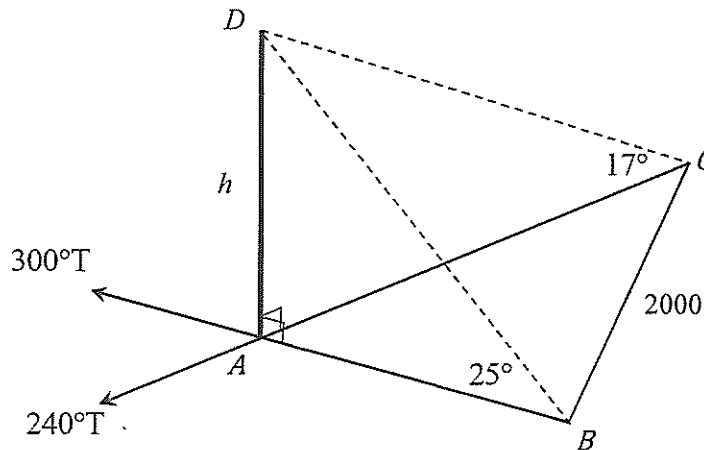
(ii) It can also be shown that:

$$\cot \theta + \tan \theta = 2 \operatorname{cosec} 2\theta. \quad (\text{DO NOT PROVE THIS})$$

Use this result and the result from part i) to find the exact value of $\cot \frac{\pi}{12}$. 2

Question 7 (11 marks) Use a SEPARATE writing booklet.

- (a) The diagram below shows Belinda at point B and Carrie at point C who are on the same horizontal level, 2000 metres apart. An aeroplane is at point D directly above the point A at an altitude of h metres. At the same time, Belinda observes the plane on a bearing of 300°T at an angle of elevation of 25° degrees and Carrie observes the plane on a bearing of 240°T at an angle of elevation of 17° .



- (i) Show that $AB = h \cot 25^\circ$. 1

- (ii) By first finding a similar expression for AC , show that: 3

$$h = \frac{2000}{\sqrt{\cot^2 25^\circ + \cot^2 17^\circ - \cot 25^\circ \cot 17^\circ}}.$$

- (iii) Using $h \approx 695$ m, find the bearing of C from B to the nearest degree. 2

- (b) (i) By expanding $\cos 2A$ show that $2 \cos^2 A = 1 + \cos 2A$. 1

- (ii) On the same set of axes, sketch the graphs of: 2

$$y = \sin x \text{ and } y = 1 + \cos x \text{ for } 0 \leq x \leq 2\pi.$$

- (iii) By using part (i) and part (ii) solve: 2

$$2 \operatorname{cosec} x \leq \sec^2 \left(\frac{x}{2} \right) \text{ for } 0 \leq x \leq \pi.$$

End of Exam

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NORTH
SYDNEY
GIRLS HIGH
SCHOOL

2018

Preliminary
Assessment
Task 2

Preliminary Mathematics Extension

SOLUTIONS

Section I

4 marks

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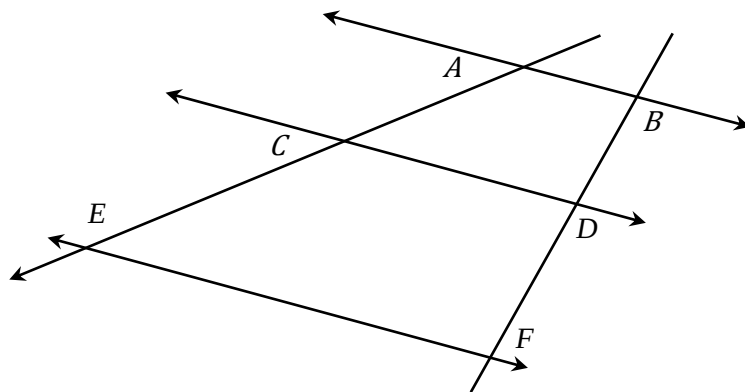
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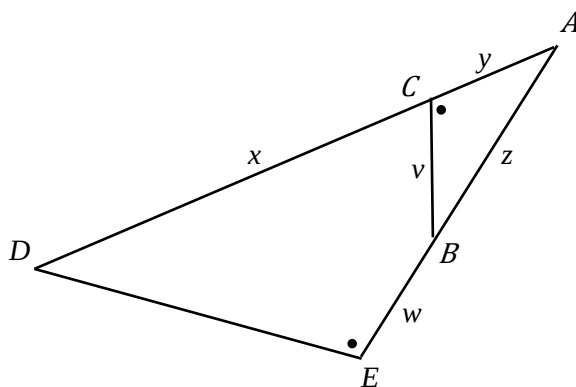
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What is the length of DE ?

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- (B) $\frac{vx + vy}{z}$
- (C) $\frac{vx + vy}{y}$
- (D) $\frac{yz + yw}{v}$

Question 5 (12 marks) Use a SEPARATE writing booklet.

(a) Prove that $\cos(A+B) - \cos(A-B) = -2\sin A \sin B$.

2

$$\begin{aligned} LHS &= \cos(A+B) - \cos(A-B) \\ &= (\cos A \cos B - \sin A \sin B) - (\cos A \cos B + \sin A \sin B) \\ &= \cos A \cos B - \sin A \sin B - \cos A \cos B - \sin A \sin B \\ &= -2\sin A \sin B \end{aligned}$$

(b) Let $\sin x + \sqrt{3} \cos x \equiv R \sin(x + \alpha)$.

(i) By expanding the right-hand side and equating coefficients, find the values of R and α .

3

$$\begin{aligned} RHS &= R \sin(x + \alpha) \\ &= R(\sin x \cos \alpha + \sin \alpha \cos x) \\ &= R \cos \alpha \sin x + R \sin \alpha \cos x \end{aligned}$$

By equating co-efficients of $\cos x$ and $\sin x$

$$1 = R \cos \alpha \quad I \quad \text{and} \quad \sqrt{3} = R \sin \alpha \quad II$$

By squaring and adding

$$1 = R^2 \cos^2 \alpha \quad I^2 \quad \quad 3 = R^2 \sin^2 \alpha \quad II^2$$

$$1 + 3 = R^2 (\cos^2 \alpha + \sin^2 \alpha)$$

$$4 = R^2$$

$$R = 2$$

$$1 = 2 \cos \alpha \quad I \quad \text{and} \quad \sqrt{3} = 2 \sin \alpha \quad II$$

$$\frac{1}{2} = \cos \alpha \quad \text{and} \quad \frac{\sqrt{3}}{2} = \sin \alpha$$

Since both $\cos \alpha$ and $\sin \alpha$ are positive then α is in the 1st quadrant

$$\alpha = \cos^{-1} \frac{1}{2}$$

$$= \frac{\pi}{3}$$

$$\sin x + \sqrt{3} \cos x \equiv 2 \sin \left(x + \frac{\pi}{3} \right)$$

(ii) Hence, or otherwise, find the maximum value of the function

1

$$f(x) = \sin x + \sqrt{3} \cos x - 5.$$

$$f(x) = \sin x + \sqrt{3} \cos x - 5$$

$$f(x) = 2 \sin \left(x + \frac{\pi}{6} \right) - 5$$

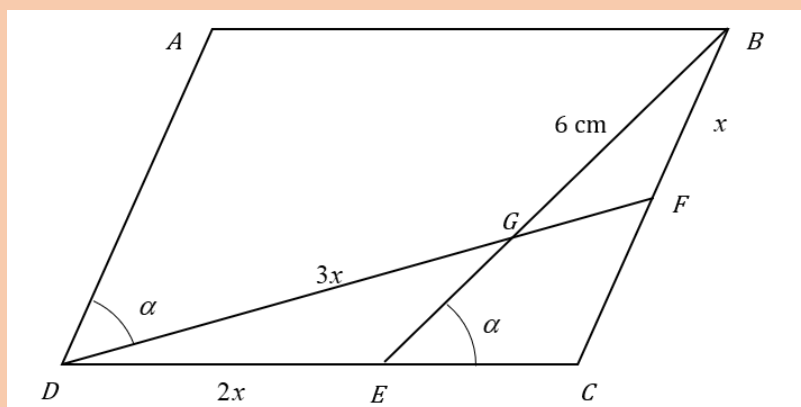
The maximum value of $2 \sin \left(x + \frac{\pi}{6} \right)$ is 2.

Maximum value of $f(x)$ is $2 - 5 = -3$

(c) $ABCD$ is a parallelogram.

F and E are two points on BC and DC respectively such that $BF = x$, $DE = 2x$ and $\angle ADF = \angle BEC = \alpha$.

FD meets BE at G such that $DG = 3x$ and $BG = 6 \text{ cm}$.



Copy the diagram into your writing booklet

(i) Prove that $\angle GDE = \angle GBF$.

2

Let $\angle GDE = \beta$

$AB \parallel DC$ (Opposite sides of a parallelogram)

$\angle CEG = \angle ABG = \alpha$ (Alternate angles on parallel lines $AB \parallel DC$)

$\angle ADE = \angle ABF$ (Opposite angles of a parallelogram)

$\angle ADG + \angle GDE = \angle ABG + \angle GBF$

$\alpha + \angle GDE = \alpha + \angle GBF$

$\angle GDE = \angle GBF$

(ii) Prove that $\triangle GDE$ is similar to $\triangle GBF$.

2

In $\triangle GDE$ and $\triangle GBF$

$$\angle DGE = \angle AGF \quad (\text{Vertically opposite angles})$$

$$\angle GDE = \angle GBF \quad (\text{Part i))}$$

$$\triangle GDE \parallel \triangle GBF \quad (\text{Equiangular})$$

(iii) Hence find the value of x .

2

$$\frac{GB}{GD} = \frac{BF}{DE} \quad (\text{Matching sides in Similar Triangles})$$

$$\frac{6}{3x} = \frac{x}{2x} \quad x \neq 0$$

$$\frac{2}{x} = \frac{1}{2}$$

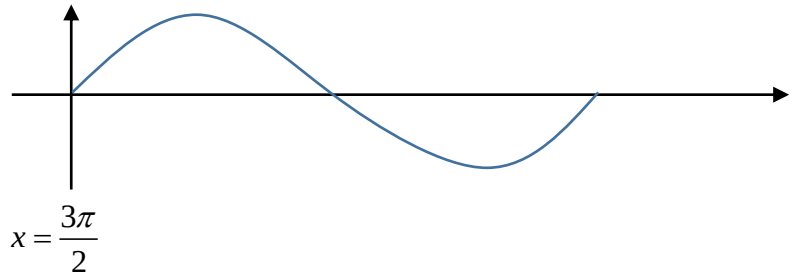
$$x = 4$$

Question 6 (13 marks) Use a SEPARATE writing booklet.

(a) Solve the equation $\cos 2\alpha = \sin \alpha$ for $0 \leq \alpha \leq 2\pi$.

3

$$\begin{aligned}\cos 2\alpha &= \sin \alpha & 0 \leq \alpha \leq 2\pi \\ 1 - 2\sin^2 \alpha &= \sin \alpha \\ 0 &= 2\sin^2 \alpha + \sin \alpha - 1 \\ 0 &= (2\sin \alpha - 1)(\sin \alpha + 1) \\ 2\sin \alpha - 1 &= 0 & \sin \alpha + 1 = 0 \\ \sin \alpha &= \frac{1}{2} & \sin \alpha = -1 \\ x \text{ is in 1}^{\text{st}} \text{ and 2}^{\text{nd}} \text{ quadrant.} \\ x &= \frac{\pi}{6} \\ x &= \pi - \frac{\pi}{6} \\ &= \frac{5\pi}{6} \\ x &= \frac{3\pi}{2}\end{aligned}$$



(b) Use the substitution $t = \tan \frac{x}{2}$ to solve

3

$$\sin x + \cos x = -1 \text{ for } 0 \leq x \leq 2\pi.$$

$$\sin x + \cos x = -1 \text{ for } 0 \leq x \leq 2\pi.$$

$$t = \tan \frac{x}{2} \text{ then}$$

$$\sin x = \frac{2t}{1+t^2} \quad \text{and} \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = -1$$

$$2t + 1 - t^2 = -1 - t^2$$

$$2t + 1 = -1$$

$$2t = -2$$

$$t = -1$$

$$\tan \frac{x}{2} = -1 \quad 0 \leq x \leq 2\pi$$

$$\tan u = -1 \quad 0 \leq u \leq \pi$$

u is in the 2nd quadrant

$$u = \pi - \frac{\pi}{4}$$

$$= \frac{3\pi}{4}$$

$$\frac{x}{2} = \frac{3\pi}{4}$$

$$x = \frac{3\pi}{2}$$

Also test $x = \pi$ in the original equation.

$$\begin{aligned}\sin \pi + \cos \pi &= (0) + (-1) \\ &= -1\end{aligned}$$

$\therefore x = \pi$ is also a solution

(c) (i) Find a general solution of $\cos 2x = \frac{1}{2}$.

2

$$2x = 2\pi n \pm \cos^{-1} \frac{1}{2}$$

$$2x = 2\pi n \pm \frac{\pi}{3}$$

$$x = \pi n \pm \frac{\pi}{6}$$

$$x = \frac{6\pi n \pm \pi}{6}$$

$$x = \frac{(6n \pm 1)\pi}{6}$$

(ii) Hence or otherwise solve $\cos 2x = \frac{1}{2}$ for $-2\pi \leq x \leq 0$.

1

$n = 0$	$x = \frac{(0 \pm 1)\pi}{6}$ $x = \frac{-\pi}{6} \quad x = \frac{\pi}{6}$
$n = -1$	$x = \frac{(-6 \pm 1)\pi}{6}$ $x = \frac{-7\pi}{6} \quad x = \frac{-5\pi}{6}$
$n = -2$	$x = \frac{(-12 \pm 1)\pi}{6}$ $x = \frac{-13\pi}{6} \quad x = \frac{-11\pi}{6}$

Thus the solutions that are in the required domain are

$$x = \frac{-\pi}{6} \quad x = \frac{-5\pi}{6} \quad x = \frac{-7\pi}{6} \quad \text{and} \quad x = \frac{-11\pi}{6}$$

(d) (i) Prove that $\cot \theta - \tan \theta = 2 \cot 2\theta$.

2

$$\begin{aligned} LHS &= \cot \theta - \tan \theta \\ &= \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \\ &= \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta} \\ &= \frac{\cos 2\theta}{\frac{1}{2} \sin 2\theta} \\ &= 2 \cot 2\theta \end{aligned}$$

(ii) It can also be shown that:

$$\cot \theta + \tan \theta = 2 \operatorname{cosec} 2\theta. \quad (\text{DO NOT PROVE THIS})$$

Use this result and the result from part i) to find the exact value of $\cot \frac{\pi}{12}$.

2

$$\cot \theta - \tan \theta = 2 \cot 2\theta \quad I$$

$$\cot \theta + \tan \theta = 2 \operatorname{cosec} 2\theta \quad II$$

$$I + II$$

$$2 \cot \theta = 2 \operatorname{cosec} 2\theta + 2 \cot 2\theta$$

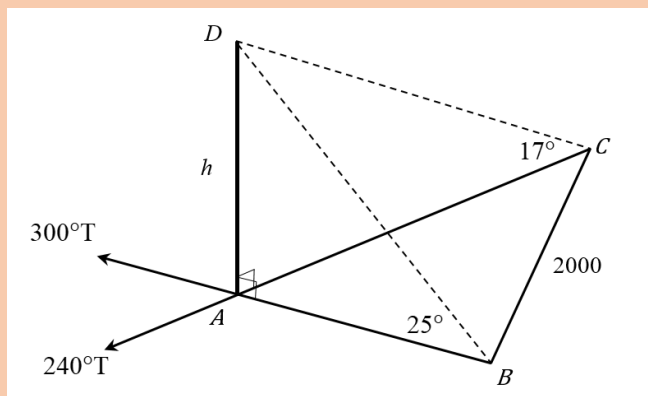
$$\cot \theta = \operatorname{cosec} 2\theta + \cot 2\theta$$

$$\text{Let } \theta =$$

$$\begin{aligned} \cot \frac{\pi}{12} &= \operatorname{cosec} \frac{2\pi}{12} + \cot \frac{2\pi}{12} \\ &= \operatorname{cosec} \frac{\pi}{6} + \cot \frac{\pi}{6} \\ &= 2 + \sqrt{3} \end{aligned}$$

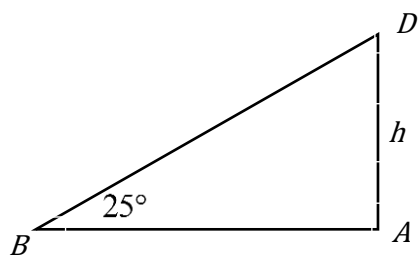
Question 7 (11 marks) Use a SEPARATE writing booklet.

- (a) The diagram below shows Belinda at point B and Carrie at point C who are on the same horizontal level, 2000 metres apart. An aeroplane is at point D directly above the point A at an altitude of h metres. At the same time, Belinda observes the plane on a bearing of 300°T at an angle of elevation of 25° degrees and Carrie observes the plane on a bearing of 240°T at an angle of elevation of 17° .



- (i) Show that $AB = h \cot 25^\circ$.

1



In $\triangle ABD$

$$\frac{AB}{AD} = \cot 25^\circ$$

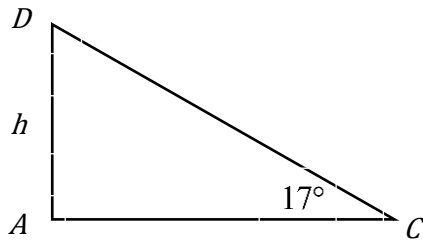
$$\frac{AB}{h} = \cot 25^\circ$$

$$AB = h \cot 25^\circ$$

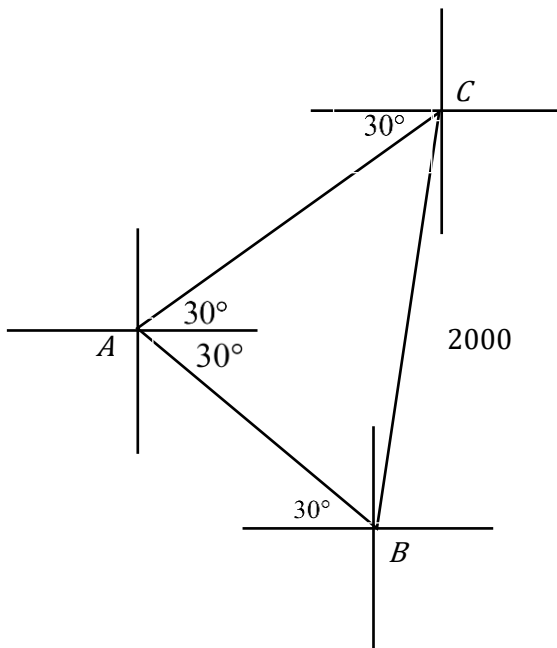
(ii) By first finding a similar expression for AC , show that:

3

$$h = \frac{2000}{\sqrt{\cot^2 25^\circ + \cot^2 17^\circ - \cot 25^\circ \cot 17^\circ}}.$$



$$AC = h \cot 17^\circ$$



In $\triangle ABC$, $\angle BAC = 60^\circ$, $BC = 2000$, $AC = h \cot 17^\circ$ and $AB = h \cot 25^\circ$

Using the cosine rule in $\triangle ABC$

$$(h \cot 25^\circ)^2 + (h \cot 17^\circ)^2 - 2(h \cot 17^\circ)(h \cot 25^\circ) \cos 60^\circ = 2000^2$$

$$h^2 \left(\cot^2 25^\circ + \cot^2 17^\circ - 2 \cot 17^\circ \cot 25^\circ \left(\frac{1}{2} \right) \right) = 2000^2$$

$$h^2 (\cot^2 25^\circ + \cot^2 17^\circ - \cot 17^\circ \cot 25^\circ) = 2000^2$$

$$h^2 = \frac{2000^2}{\cot^2 25^\circ + \cot^2 17^\circ - \cot 17^\circ \cot 25^\circ}$$

$$h = \frac{2000}{\sqrt{\cot^2 25^\circ + \cot^2 17^\circ - \cot 17^\circ \cot 25^\circ}}$$

(iii) Using $h \approx 695$ m, find the bearing of C from B to the nearest degree.

2

Method 1

Let $\angle ABC = \theta$ and $h \approx 695$

Using the cosine rule in $\triangle ABC$

$$\cos \theta = \frac{2000^2 + (695 \cot 25^\circ)^2 - (695 \cot 17^\circ)^2}{2(2000)(695 \cot 25^\circ)}$$

$$= 0.1769796411\dots$$

$$\theta = 79^\circ 48'$$

$$\theta \approx 80$$

Bearing of C from B is equal to $300 + 80 = 380 = 020^\circ T$

Method 2

Let $\angle ACB = \theta$ and $h \approx 695$

Using the sine rule in $\triangle ABC$

$$\frac{\sin \theta}{h \cot 25^\circ} = \frac{\sin 60^\circ}{2000}$$

$$\sin \theta = \frac{695 \cot 25^\circ \sin 60^\circ}{2000}$$

$$\theta = \sin^{-1} \left(\frac{695 \cot 25^\circ \sin 60^\circ}{2000} \right)$$

$$\theta = 40^\circ$$

$$\angle ABC = 180 - 60 - 40 = 80^\circ$$

Bearing of C from B is equal to $300 + 80 = 380 = 020^\circ T$

Method 3

Let $\angle ABC = \theta$ and $h \approx 695$

Using the sine rule in $\triangle ABC$

$$\frac{\sin \theta}{695 \cot 17^\circ} = \frac{\sin 60^\circ}{2000}$$

$$\sin \theta = \frac{695 \cot 17^\circ \sin 60^\circ}{2000}$$

$$\theta = \sin^{-1} \left(\frac{695 \cot 17^\circ \sin 60^\circ}{2000} \right)$$

$$\theta = 79^\circ 50'$$

$$\theta = 180 - \sin^{-1} \left(\frac{695 \cot 25^\circ \sin 60^\circ}{2000} \right)$$

$$\theta = 100^\circ 10'$$

Both these angles are possible in this triangle so this is an ambiguous case and either method 1 or method 2 must be used.

(b) (i) By expanding $\cos 2A$ show that $2\cos^2 A = 1 + \cos 2A$.

1

$$RHS = 1 + \cos 2A$$

$$= 1 + \cos^2 A - \sin^2 A$$

$$= 1 + \cos^2 A - (1 - \cos^2 A)$$

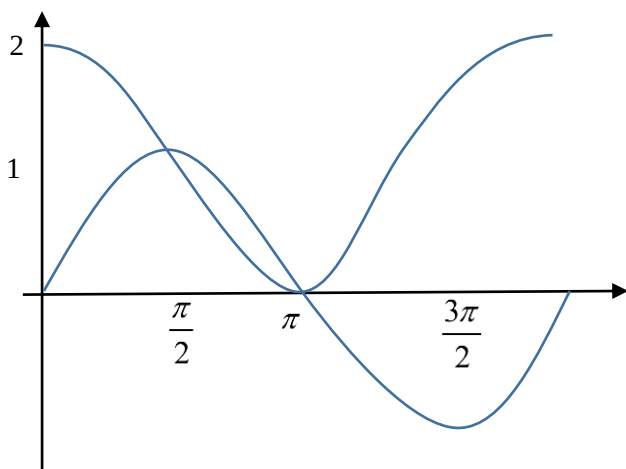
$$= 2\cos^2 A$$

$$= LHS$$

(ii) On the same set of axes, sketch the graphs of:

2

$$y = \sin x \text{ and } y = 1 + \cos x \text{ for } 0 \leq x \leq 2\pi.$$



(iii) By using part (i) and part (ii) solve:

2

$$2 \operatorname{cosec} x \leq \sec^2 \left(\frac{x}{2} \right) \text{ for } 0 \leq x \leq \pi.$$

$$2 \operatorname{cosec} x \leq \sec^2 \left(\frac{x}{2} \right)$$

$$\frac{2}{\sin x} \leq \frac{1}{\cos^2 \left(\frac{x}{2} \right)}$$

$$\frac{1}{\sin x} \leq \frac{1}{2 \cos^2 \left(\frac{x}{2} \right)}$$

$$\frac{1}{\sin x} \leq \frac{1}{1 + \cos x}$$

From Part i)

$$1 + \cos x \leq \sin x$$

Since both $\sin x > 0$ and $1 + \cos x > 0$ for $0 \leq x \leq \pi$.

Thus by considering when the graph of $y = \sin x$ is higher than $y = 1 + \cos x$ for $0 \leq x \leq \pi$

Solution is $\frac{\pi}{2} \leq x < \pi$

NOTE: $x \neq \pi$ since $\operatorname{cosec} x$ is undefined for $x = \pi$

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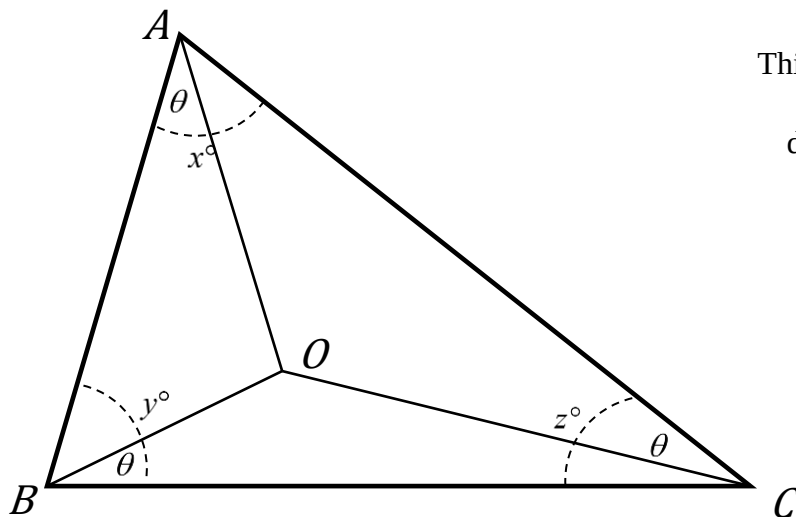
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(b) Prove the identity $\cot \alpha - \cot \beta = \frac{\sin(\beta - \alpha)}{\sin \alpha \sin \beta}$

2

(c) In the $\triangle ABC$ below, $\angle CAB = x$, $\angle ABC = y$ and $\angle ACB = z$.

The point O is placed inside $\triangle ABC$ so that $\angle AOB = \angle BOC = \angle COA = \theta$



This is **NOT**
a 3-D
diagram

(i) Show that $\frac{OX}{OY} = \frac{\sin(y - \theta)}{\sin \theta}$

2

(ii) Hence show that $\sin^3 \theta = \sin(x - \theta) \sin(y - \theta) \sin(z - \theta)$

2

(iii) Hence, using parts (c) (ii) and (b) show that:

2

$$(\cot \theta - \cot x)(\cot \theta - \cot y)(\cot \theta - \cot z) = \operatorname{cosec} x \operatorname{cosec} y \operatorname{cosec} z .$$

End of paper